Theorems, proofs, and lemmas in *Cops, Robbers, and Barricades* — Meger (2016)

An aggregation of central theorems, lemmas, and proofs from Dr. Meger's 2016 Master of Applied Mathematics thesis, separated by chapter and supplemented with proof outlines + page references. For full proofs, read the paper here.

condensed theorems (summary list)

Theorem 1: given a connected graph $G = (VE)$ with $|V(G)| = n$, the following are equivalent: G is a tree, G has $n - 1$ edges, there is a unique path between any pair of vertices $u, v \epsilon V(G)$.

Theorem 2: for a graph G of order n, $\gamma(G) \leq n - \Delta(G)$.

Theorem 3: If for every $x, y \epsilon V(G)$ $x \leq y$, then G is cop-win.

Theorem 4.1: For a graph G, we have that $C(G) \leq \gamma(G)$.

4.2: If G is a tree, then $C(G) = 1$.

4.3: For $n \geq 4$ and integer, $C(C_n) = 2$.

Lemma 5: For any graph G, $c(G) \leq C_B(G)$ where $C_B(G)$ is the min number of cops required to guarantee capture in CRB and $C(G)$ is the min number of cops required to guarantee capture in standard cops and robbers (no barricades).

Theorem 6: For any graph G, $C_B(G) \leq \gamma(G)$ where $\gamma(G)$ is the domination number of G.

Theorem 7: For a path with $n \epsilon N$ vertices P_n, we have that $C_B(P_n) = ceiling(\frac{n+1}{6})$.

Theorem 8: For a cycle with $n \geq 3 \epsilon N$ vertices C_n, we have that $C_B(C_n) = ceiling(\frac{n}{6})$.

Theorem 9: If there exists a robber-star, then we require a cop on the path between x and each star vertex.

Theorem 10: For a tree T of height ≥ 2 with set of leaves l, we have that $C_B(T) \geq |N_2(l)|$.

Theorem 11: A tree is barricade-cop-win if there exists a vertex v st when T is rooted at v, T has a height of ≤ 2.

Theorem 12: If T is a complete m-ary tree with height h,$C_B(T) \leq \sum_{i=0}^{c} m^{(c-i)(m+4)+m+1}$ where $c = ceiling(\frac{h-m-3}{m+4})$.

Theorem 13: If the cops are forced to play the row strategy (see page 30), then we have that $C_B(T) = \sum_{i=0}^{c} m^{(c-i)(2m+4+m+1}$, where $c = ceiling(\frac{h-m-3}{2m+4})$.

Theorem 14: If a tree T has height h and a maximum degree of $m + 1$, then $C_B(T) \leq \sum_{i=0}^{c} m^{(c-i)(2m+4)+m+1}$ where $c = ceiling(\frac{h-m-3}{2m+4})$.

theorem 15: For a locally path-like graph G,

$C_B(G) \leq \sum_{e\epsilon E(W)} ceiling(\frac{w(e)}{6} - 1) + |V(W(G))|$ where $w(e)$ is the weight of an edge e in $W(G)$.

Theorem 16: If a graph G with $V(G) = n$ and no isolated vertices, then $\gamma_2(G) \leq \frac{n-\Delta+1}{2}$.

Theorem 17: If a graph G with $V(G) = n$ has girth $\geq$ 7 then $C(G) \leq \gamma_2$ and $C_B(G) \leq \gamma_2$.

Theorem 18: If G has $\Delta \geq (\sqrt{n} - 1)^2$ then $\gamma_2 \leq \sqrt{n}$.

Corollary 19: If G has girth $\geq$ 7 and $\Delta \geq (\sqrt{n} - 1)^2$, then $C(G) \leq \sqrt{n}$.

Theorem 20: Given a graph G with a particular starting placement of cops and the robber st ∃ a robber cut of size k. If the minimum distance between all cops and any vertex in the cut is strictly > k, then the robber wins when playing with global barricade rules.

Theorem 21: For any graph G, $C(G) \leq C^3{}_B(G) \leq C_B(G) = C^2{}_B(G) \leq C^1{}_B(G)$.*

Theorem 25: A graph G is barricade-cop-win iff for every play of the game, for all rounds $k \geq 1$ with corresponding barricade sequence B_k, there is some vertex $u \epsilon N_{G_{k+1}}(C)$ st $\epsilon\, comp_{G_k}(R)$ for any k st $u \leq_\rho R$ for some finite ρ.

proof sketches

chapter 1, pages 1-19

Theorem 1

Given a connected graph $G = (VE)$ with $|V(G)| = n$, the following are equivalent:

- G is a tree
- G has $n - 1$ edges
- There is a unique path between any pair of vertices $u, v \epsilon V(G)$.

Theorem 2

For a graph G of order n, we have that $\gamma(G) \leq n - \Delta(G)$.

➔ Add every vertex in $V(G)$ to a dominating set S.
➔ Remove all neighbours of some vertex v from S.
➔ The only vertices not in S are the ones adjacent to v.
➔ For any dominating set S, $\gamma(G) \leq |S|$
➔ So QED

Theorem 3

If for every $x, y \epsilon V(G)$ $x \leq y$, then G is cop-win.

➔ No matter what starting position is chosen, the cop can choose a vertex so that wherever the robber chooses to move, the cop will be closer to the robber than when he was previously.
➔ The cop will capture the robber.

Theorem 4

1. For a graph G, we have that $C(G) \leq \gamma(G)$.
2. If G is a tree, then $C(G) = 1$.
3. For $n \geq 4$ and integer, $C(C_n) = 2$.

➔ For item 1, note that every vertex in a dominating set is adjacent to every vertex outside the set. If the cops play on all vertices of a dominating set, then they will capture the robber on their first move.
➔ For item 2, note that for a tree there exists a unique path between any pair of vertices u, v ∈ V (G). The cop follows the unique path between himself and the robber. This path will never increase in length, and will eventually decrease until the robber is captured.
➔ For item 3, observe that for a cycle, if a cop attempts to follow the robber, then they will simply go around the cycle in circles indefinitely. Thus, we

need one cop to chase the robber in one direction, and a second cop to come around the cycle in other direction, eventually closing

➔ in on the robber from both sides.

Lemma 5

For any graph G, $C(G) \leq C_B(G)$ where $C_B(G)$ is the min number of cops required to guarantee capture in CRB and $C(G)$ is the min number of cops required to guarantee capture in standard cops and robbers (no barricades).

- ➔ $C_B(G) \geq 1$ because 0 cops will always lose (base case)
- ➔ For any graph G, if G is $c(G) \geq 2$ and has $(c(G) - 1))$ cops then we can play CRB.
- ➔ Need at least $c(G)$ cops to capture the robber.

Giving the robber more options can only help the robber, not hurt them. Any number of cops that fails in the standard game will still fail if you let the robber barricade.

Theorem 6

For any graph G, $C_B(G) \leq \gamma(G)$ where $\gamma(G)$ is the domination number of G (the least size of a vertex set st every other vertex in the graph has a neighbour in it, via Petr et al., 2022).

- ➔ The set of cops play on all the vertices of a dominating set of minimum size. For any initial placement of the robber, a cop will be able to capture R in the first round.

Theorem 7

Consider the path on 8 vertices. Notice that $c(P_8) = 1$. If one cop is placed on the middle vertex, the robber will go to an end-vertex and place a barricade between herself and the cop. The robber wins. If instead there are two cops at each endpoint (denoted v_1 and v_8), the robber can place herself close to the middle (v_4) and place barricades to her left and right, preventing both cops from reaching her and winning the game.

For a path with $n \epsilon N$ vertices P_n, we have that $C_B(P_n) = ceiling(\frac{n+1}{6})$.

- ➔ Label the vertices of P_n. Place a cop on v_3, who can guard $v_1 \wedge v_2$.
- ➔ A cop starting at v_4 would not be able to guard v_1 because R can build a barricade on v_2 while the cop is at v_3.
- ➔ If we placed a cop on every sixth vertex, the cops have a winning strategy.

➔ Fewer cops would allow the robber to win.

Theorem 8

For a cycle with $n \geq 3 \epsilon N$ vertices C_n, we have that $C_B(C_n) = ceiling(\frac{n}{6})$.

➔ Place a cop at some vertex v_1.

➔ Continue around the cycle placing a cop on every sixth vertex.

➔ At the end we have at most five vertices between the last placed cop and the cop on v_1. This results in a winning strategy for the cops.

➔ The robber can build herself into a win if the distance between some pair of cops is greater than five.

Theorem 9

If there exists a robber-star, then we require a cop on the path between x and each star vertex.

➔ Suppose there is some star vertex u that does not have a cop on the path between x and u.

➔ The robber will play on x and build barricades on each neighbour.

➔ Since there is no cop on u, on the deg(x)th round, the robber will be able to build a barricade on vertex v and win the game.

Theorem 10

For a tree T of height ≥ 2 with set of leaves l, we have that $C_B(T) \geq |N_2(l)|$.

➔ All grandparents must be covered by cops in order to guard leaves. If one leaf is not guarded, then the robber will place herself on that vertex. In the second round, the robber can build and win as cops get closer.

Theorem 11

A tree is barricade-cop-win if there exists a vertex v st when T is rooted at v, T has a height of ≤ 2.

➔ For a tree of height 2, we place a cop on the root of the tree.

➔ Robber will place herself on a leaf.

➔ Cop will move down the unique path between himself and the robber.

➔ The robber will have no place to move or build a barricade that round and will need to stay put. The cop will capture the robber in the following round.

- ➔ If a tree is barricade-cop-win then we know it can only have one grandparent.
- ➔ This grandparent is then a root for the tree st the tree has a height 2.

Theorem 12

If T is a complete m-ary tree with height h, then we have that

$C_B(T) \leq \sum_{i=0}^{c} m^{(c-i)(m+4)+m+1}$ where $c = ceiling(\frac{h-m-3}{m+4})$.

- ➔ We use "the row strategy".
- ➔ Place cops to occupy all vertices of a given height. They will move up and down the tree as a whole.
- ➔ Place cops at all vertices at height $h - 2$, of which there are m^{h-2}.
- ➔ The only move the robber can make is to stay still, no matter the case.

Theorem 13

If the cops are forced to play the row strategy (see page 30), then we have that $C_B(T) = \sum_{i=0}^{c} m^{(c-i)(2m+4)+m+1}$, where $c = ceiling(\frac{h-m-3}{2m+4})$.

- ➔ If we increase the spacing between cops, the robber will be able to win.
- ➔ If there is at least one cop on the child of the robber vertex after the mth move (where m is the numbre of cops) then the cops can win. Special case?

Theorem 14

If a tree T has height h and a maximum degree of $m + 1$, then

$C_B(T) \leq \sum_{i=0}^{c} m^{(c-i)(2m+4)+m+1}$ where $c = ceiling(\frac{h-m-3}{2m+4})$.

- ➔ Play the row strategy as above.
- ➔ Add multiple cops to vertices in the same height.
- ➔ This will result in a win for analogous reasons.

Theorem 15

For a locally path-like graph G we have that

$C_B(G) \leq \sum_{e \epsilon E(W)} ceiling(\frac{w(e)}{6}) - 1 + |V(W(G))|$ where $w(e)$ denotes the weight of an edge e in $W(G)$.

- ➔ Place cops on all vertices in $W(G)$.

- ➔ $\forall\{u, v\}\epsilon W(G)$, they are paths in our original graph G.
- ➔ The first six vertices are guarded by the cop on u; thus they have a path of length $w(\{u, v\})$ to guard.
- ➔ we must guard every 6th vertex.
- ➔ the number of cops we require is $ceiling(\frac{w(\{u,v\})}{6} - 1)$.
- ➔ for each loop $\{u, u\}$ we use the same strategy. final vertices guarded by u as well.

<u>Theorem 16</u>

If a graph G with $V(G) = n$ and no isolated vertices, then $\gamma_2(G) \leq \frac{n-\Delta+1}{2}$.

<u>Theorem 17</u>

If a graph G with $V(G) = n$ has girth ≥ 7 then $C(G) \leq \gamma_2$ and $C_B(G) \leq \gamma_2$.

- ➔ Consider the figure on page 38.
- ➔ Place cops on each vertex of a distance 2 dominating set of minimum size.
- ➔ The robber is dominated by cops at most distance 2 and all her neighbours are dominated by cops at most distance 2.
- ➔ First move of cops will all move on the unique shortest paths connecting themselves and the robber, using that the girth is at least 7.
- ➔ if the robber moves to any neighbour she will be adjacent to a cop and therefore will be captured in the second round.

<u>Theorem 18</u>

If G has $\Delta \geq (\sqrt{n} - 1)^2$ then $\gamma_2 \leq \sqrt{n}$.

- ➔ Consider the derivation on page 39 which follows from the bound in theorem 13.

<u>Corollary 19</u>

If G has girth $\geq$ 7 and $\Delta \geq (\sqrt{n} - 1)^2$, then $C(G) \leq \sqrt{n}$.

- ➔ Proof follows by theorems 14 and 15.

chapter 3, pages 42-52

Theorem 20

Given a graph G with a particular starting placement of cops and the robber st ∃ a robber cut of size k. If the minimum distance between all cops and any vertex in the cut is strictly > k, then the robber wins when playing with global barricade rules.

- ➔ Cops must eventually cross the robber cut to capture the robber therefore on their first move, at least one cop will move toward the cut.
- ➔ In her first move, the robber will lay a barricade on one vertex of the cut.
- ➔ The distance from any cop to the cut is >k so on their k^{th} turn no cop will be on a vertex in the cut. On her final turn the last vertex is unoccupied and she builds herself in, winning the game.
- ➔ There are $n \cdot C(n, k)$ — where C is combination; google docs has limited math symbols lol — starting positions when we play with k cops.
- ➔ the number of starting arrangements grows quickly from n^2.

Theorem 21

For any graph G, we have that

$C(G) \leq C^3_B(G) \leq C_B(G) = C^2_B(G) \leq C^1_B(G)$.

Lemma 22

For any graph G we have that $C(G) \leq C^3_B(G)$.

- ➔ We require at least $C(G)$-many cops when playing the slug barricade rules.
- ➔ Otherwise the robber would play her winning strategy in the original game.

Lemma 23

For any graph G, we have that $C^3_B(G) \leq C^2_B(G)$.

- ➔ We play the local game using $C^3_B(G) - 1$ cops.
- ➔ The robber can still use the slug strategy.
- ➔ Since there are less than $C^3_B(G)$ many cops, the robber will be able to win.

Lemma 24

For any graph G, we have that ${C^2}_B(G) \leq {C^1}_B(G)$.

➔ The robber can use her winning strategy from the local game in order to win in the global game.

Further proof of theorem 21 on page 49.

chapter 4, pages 52-63

Theorem 25

A graph G is barricade-cop-win iff for every play of the game, for all rounds $k \geq 1$ with corresponding barricade sequence B_k, there is some vertex $u \epsilon N_{G_{k+1}}(C)$ st $\epsilon\ comp_{G_k}(R)$ for any k st $u \leq_{\rho} R$ for some finite ρ.

- ➔ Assume G is barricade-cop-win.
- ➔ G must be cop-win in the original game.
- ➔ In round 0, $R \leq_{\rho} C$ for some ρ.
- ➔ that cop is in the same connected component as the robber.
- ➔ Assume the result holds for some $k \geq 0$:
  - ◆ in the kth round consider the cop's move.
  - ◆ he will eventually capture the robber.
  - ◆ there must be some vertex $v \epsilon N_{G_{k-1}}(C)$ that is in the same component of R that is not also in B_k.
  - ◆ Since the cop eventually catches the robber, there exists some finite r st $u \leq_{r} v$ in some G_k as required.
- ➔ If C takes a combination of the two types of moves listed, C will capture R by moving into $comp_{G_{k'}}(R)$ for some k' via R, or C will follow the strategy using the relations to capture the robber R within its component.

note: barricade-cop-win algorithm on page 58.

Open problems

- We conjecture that the row strategy is the optimal strategy for the cops when playing on m-ary trees. The upper bound on cB we proved for trees is likely not tight, since we currently use the bound for m-ary trees as an upper bound for all trees. Can we utilize graph packings to determine a lower bound for general trees?
- In the version of the CRB game where we play with the slug variant, we would like to consider the possibility that the robber must build a barricade on every turn. This would imply the robber can never return to a previously visited vertex. How would this affect the barricade-cop number?
- There is much to investigate in regards to the complexity of the algorithm presented in Chapter 4. What is the exact time complexity of computing cB? Would it be possible to determine if a graph is barricade-cop-win in polynomial time, or is the problem NP-hard?
- Can we extend the barricade-cop-win characterization in Chapter 4 to the case of k-barricade-cop-win graphs, where $k > 1$?
- A question that was not touched in the thesis would be to add randomness to this game. What is the barricade-cop number on the binomial random graph $G(n, p)$? When playing on deterministic graphs, what happens if the barricades are placed by the robber at random?
- It would be interesting to play the CRB game on infinite graphs. For which graphs do we require an infinite number of cops? Is it possible for the robber to build enough barricades to reduce gameplay to a finite graph?